

Abhay Sharma

Chennai, India | +91 83198 81793 | 24f1001915@ds.study.iitm.ac.in

github.com/git-of-abhay | linkedin.com/in/abhaysharma12 | x.com/letmecodeee | abhaysharma.dev

Experience

Ecowell India

Remote

Software Development Engineer Intern

April 2026 – September 2026

- Built and launched a customized CRM supporting 10,000+ customers across 8 locations, coordinating operational workflows across 18 teams and 8 role-based access levels.
- Developed cross-platform web and mobile applications using JavaScript, TypeScript, React, and React Native, backed by Node.js, Express.js, PostgreSQL, and SQL.
- Integrated 48 API endpoints supporting 1,000+ daily tickets and more than 10,000 transactions across sales, service, attendance, payment, and logistics operations.
- Designed GPS-based attendance and real-time workforce tracking for 1,000+ employees using mapping APIs, role-aware visibility, and location-driven business rules.
- Automated 70+ multi-stage workflows while managing frontend, backend, database, and cross-platform dependencies within a three-person engineering team operating without senior technical supervision.

Projects

CF Insights

cfinsights.netlify.app

React, TypeScript, JavaScript, HTML, CSS, Codeforces API

github.com/git-of-abhay/cfinsights

- Independently designed, developed, and deployed a Codeforces analytics platform used by 20,000+ users, with approximately 200 daily active users.
- Integrated Codeforces APIs to analyze ratings, contests, submissions, problem-solving patterns, and competitive-programming performance.
- Generated personalized performance insights and improvement plans to help users identify weak topics and make data-driven practice decisions.
- Built the frontend to handle peak traffic exceeding 100 requests per second while maintaining a responsive analytics experience.

Relief Net

github.com/Git-of-abhay/relief-net

Python, FastAPI, WebSockets, SQLite, JavaScript, HTML, CSS, PWA

- Built an offline-first progressive web application enabling real-time communication between devices on the same local network without internet access or cloud infrastructure.
- Designed 7 REST endpoints and 1 WebSocket route for room-based messaging, presence, history, moderation, health monitoring, and CSV exports.
- Implemented a persistent offline queue with automatic 2-second reconnection, service-worker caching, and configurable 30-day message retention.
- Added bounded 200-message history, rate limiting, origin validation, role-protected moderation, and audit logging across 3 SQLite data entities.
- Verified messaging, WebSocket delivery, offline persistence, security, moderation, exports, and burst handling through 8 passing automated tests.

LRU Cache – C++20

github.com/Git-of-abhay/lru-cache-cpp

C++20, STL, Templates, Multithreading, CMake, GitHub Actions

- Implemented a reusable LRU cache with average $O(1)$ lookup, insertion, update, and eviction, including an interactive CLI and thread-safe wrapper.
- Validated correctness through 100,000 randomized reference-model operations, edge and collision tests, exception rollback checks, and an 8-thread stress test covering 40,000 reads.
- Benchmarked 200,000-operation seeded workloads, measuring up to $57.6\times$ faster request processing than a linear vector baseline, with 4 GCC/Clang CI configurations including sanitizer checks.

Technical Skills

Languages: C++, Python, JavaScript, TypeScript, SQL

Frameworks: React, React Native, Node.js, Express.js, Django, Flask, FastAPI

Data and Databases: PostgreSQL, MySQL, SQLite, NumPy, pandas, scikit-learn, Matplotlib, Seaborn

Tools and Systems: Git, GitHub, Linux, CMake, GitHub Actions, REST APIs, WebSockets, pytest, Ruff, PWA

Education

Indian Institute of Technology Madras

2024 – 2028

Bachelor of Science in Data Science and Applications